

반도체공정응용(공)(반) (SM306-01)

학기	2026년 2학기
담당교수	권혁준
이수구분	심화필수
학점	3.0
강의시간	월09:00-10:30(E7 - L20), 수09:00-10:30(E7 - L20)
강의실	E3 609 (교수실)
소속	반도체공학과

교과개요

- This course introduces the semiconductor fabrication processes used in modern semiconductor industries, covering advanced logic devices such as FinFET and Gate-All-Around (GAA) transistors, as well as memory devices including DRAM and 3D NAND flash memory. Students will learn the core manufacturing steps required to fabricate these devices, including deposition, lithography, etching, doping, metallization, and integration processes. The course emphasizes the relationship between device structure and process technology, helping students understand how advanced semiconductor devices are manufactured in industrial environments.

교과목표

- Understand the fundamental semiconductor fabrication processes used in the semiconductor industry. - Explain the process flow for advanced logic devices such as FinFET and GAA transistors. - Describe the manufacturing processes of memory devices including DRAM and 3D NAND flash. - Analyze the relationship between device architecture and process integration. - Identify the technological challenges in advanced semiconductor manufacturing. - Apply semiconductor process knowledge to understand current industrial fabrication trends.

주차별 강의계획

1주차: Overview of semiconductor fabrication process
2주차: Essential background for fabrication and device
3주차: Introduction to Semiconductor Industry and Device Trends
4주차: Logic (1): the Well & STI
5주차: Logic (2): Transistor
6주차: Logic (3): Replacement Gate
7주차: Logic (4): Contact & Metallization
8주차: Mid-term exam
9주차: Introduction to Flash & DRAM memory
10주차: NAND (1): Memory stack formation
11주차: NAND (2): Word line, Bit line
12주차: NAND (3): Interconnect
13주차: DRAM (1): Transistor
14주차: DRAM (2): Capacitor
15주차: DRAM (3): Metallization
16주차: Final exam

수업방법

교과정책(방침)

- Credit for this course is not permitted if EE404 in the Electrical and Electronic track has already been taken. -
- Conversely, if you take this course, you will not be allowed to register for EE404 in the Electrical and Electronic track. - Please note that attendance will not be acknowledged in the event of absence without prior notification at least one hour before the class begins.